Binomial Distribution

1 Definition

A discrete random variable X is said to have a Binomial Distribution if its PMF is given by,

$$f_X(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, \ldots, n \\ 0, & \text{otherwise} \end{cases}, n \in \mathbb{N} \text{ and } 0 < p < 1, q = 1 - p$$

Remark

Binomial Expansion -

$$\sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x} = (p+q)^n \quad n \in \mathbb{N}$$

For, $p \geqslant 0, q = 1 - p \geqslant 0, n \in \mathbb{N}$

$$\binom{n}{x} p^x q^{n-x} \geqslant 0, \quad x = 0(1)n$$

and

$$\sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x} = 1$$

If $p = 0, \quad f_X(0) = 1$ i.e. $P[X = 0] = 1$
If $p = 1, \quad f_X(x) = 1$ i.e. $P[X = x] = 1$
If $p = 0$, or 1 X becomes a Degenerate Random variable.

2 Moments

$$\begin{aligned} \mu_{(r)} = E(X_r) &= \sum_{x=0}^{n} (x)_r \binom{n}{x} p^x q^{n-x} \\ &= \sum_{x=r}^{n} \frac{n}{(n-x)!(x-r)!} p^x q^{n-x} \\ &= (n)_r \sum \frac{(n-r)!}{(x-r)!(n-x)!} p^{x-r} q^{n-r} \\ &= (n)_r p^r \sum_{y=x-r=0}^{n-r} \frac{(n-r)!}{(\underbrace{x-r}_{y})!(n-x)!} p^{x-r} q^{n-r-(x-r)} \\ &= (n)_r p^r \sum_{y=0}^{n-r} \binom{n-r}{y} p^y q^{n-r-y} \\ &= (n)_r p^r \cdot 1, r = 1, 2, \ldots, n. \end{aligned}$$

and $\mu_{(r)} = 0$ if $r > n$
$E(X) = np$

$$
\begin{aligned}
V(X) &= E\left(X^2\right) - E^2(X) = E[X(X-1)] + E[X] - E^2(X) \\
&= n(n-1)p^2 + np - (np)^2 \\
&= np(1-p) = npq \\
\therefore E(x) &= np > npq = \mathrm{Var}(X)
\end{aligned}
$$

1. If $X \sim \mathrm{Bin}(n,p)$, show that $E\left(X^k\right) = npE(Y+1)^{k-1}$ where $Y \sim \mathrm{Bin}(n-1,p)$. Hence, find $E(X)$ and $\mathrm{Var}(X)$.

$$
\begin{aligned}
E\left(X^k\right) &= \sum_{x=0}^{n} x^k \binom{n}{x} p^x q^{n-x} \\
&= \sum_{x=1}^{n} x^{k-1} \frac{n(n-1)!}{(x-1)!(n-x)!} p^x q^{n-x} \\
&= np \sum_{x=1}^{n} x^{k-1} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x} \\
&= np \sum_{x=1}^{n} x^{k-1} \binom{n-1}{x-1} p^{x-1} q^{n-1-(x-1)} \\
&= np \sum_{y=x-1=0}^{n-1} (y+1)^{k-1} \quad \binom{n-1}{y} p^y q^{n-1-y} \\
&= npE(Y+1)^{k-1}, \quad Y \sim \mathrm{Bin}(n-1,p)
\end{aligned}
$$

For $k=1$, $E(X) = np \cdot E(Y+1)^0 = np$
For $k=2$,

$$
\begin{aligned}
E\left(X^2\right) &= npE(Y+1) = np\{E(Y)+1\} \\
&= np\{(n-1)p+1\} = \left\{n(n-1)p^2 + np\right\} \\
\mathrm{Var}(x) &= E\left(X^2\right) - E^2(X) = np(1-p) = npq
\end{aligned}
$$

3 Probability Model

Consider a series of n Bernoulli trails where n is a prefixed positive integer. Let p be the probability of success in each trail. If X denotes the no of success in n trails,

$$X \sim \mathrm{Bin}(n,p)$$

Justification

Let x be any mass point of X,

$$P[X=x] = P \text{ [there are } x \text{ success]}$$

To calculate $P[X=x]$, let us consider a specific pattern of outcomes in which first x trails results in a success, the last $(n-x)$ trails results in a failure. Such a pattern can be realized in $\binom{n}{x}$ ways. If A_i be the event that ith trail results in a success, A_i's are independent. $P(A_i) = p \quad \forall i = 1, \ldots, n$,

$$
\begin{aligned}
p[X=x] &= \binom{n}{x} p\left[A_1, A_2, \ldots A_n, A_{n+1}^c, A_{n+2}^c, \ldots A_n^c\right] \\
&= \binom{n}{x} p^x q^{n-x}, \quad x = 0,1,2,\ldots n. \quad (\text{Since} A_i\text{'s are independent})
\end{aligned}
$$

2. If $X_1, X_2, X_3 \ldots, X_n$ are i.i.d, Bernoulli (p) R.V's, show that $Y = \sum_{i=1}^{n} X_i \sim \mathrm{Bin}(n,p)$.

$P[Y=k] = P\left[\sum_{i=1}^{n} X_i = k\right] =$ Probability that k of the X_i's are 1 and $(n-k)$ of the X_i's are 0.

$$= \binom{n}{k} p\left[X_1 = 1, X_2 = 1, \ldots, X_k = 1, X_{k+1} = 0, \ldots X_n = 0\right]$$
$$= \binom{n}{k} p\left[X_1 = 1\right]^k p\left[X_{k+1} = 0\right]^{n-k}$$
$$= \binom{n}{k} p^k q^{n-k} \quad [\text{ As X's are iid }] \quad k = 0, \ldots, n$$

Hence $Y = \sum_{i=1}^{k} X_i \sim \text{Bin}(n, p)$.

3. Let mutually independent Random variable X_1, X_2, X_3 have the same pdf.

$$f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the probability that two of these variables exceed 1/2.

Let x be an observation from the p.d.f $f(x) = \begin{cases} 3x^2 & 0 < x < 1 \\ 0 & ow \end{cases}$ and $x > 1/2$ (consider) as a success.

Then, three independent R.V's can be treated as 3 independent Bernoulli trail with p (success) $= p = p[x > 1/2] = \int_{1/2}^{1} 3x^2 dx = \frac{7}{8}$.

Let Y be the no of X_i 's which are $> 1/2$ i.e. the no of successes in 3 independent Bernoulli trails.

$$Y \sim \text{Bin}(n = 3, p = 7/8)$$

$\therefore$ Required probability $P[Y = 2] = \binom{3}{2} \left(\frac{7}{8}\right)^2 \left(\frac{1}{8}\right)^1$

Alternative,

The probability that two of X_1, X_2, X_3 exceed $1/2 = \binom{3}{2} P\left[X_1 > \frac{1}{2}, X_2 > \frac{1}{2},\ X_3 < \frac{1}{2}\right]$ two X are which will exceed 1/2 can be chosen in $\binom{3}{2}$ ways and each of the cases has probability $P\left[X_1 > \frac{1}{2}, X_2 > \frac{1}{2}, X_3 < \frac{1}{2}\right]$

$$= \binom{3}{2} P\left[X_1 > \frac{1}{2}\right]^2 P\left[X_3 < \frac{1}{2}\right]$$
$$= \binom{3}{2} \left(\frac{7}{8}\right)^2 \left(\frac{1}{8}\right)$$

4. A drunk performs a Random walk over positions $0, \pm 1, \pm 2, \ldots$ as follows. He starts at zero, he takes successive one unit steps going to the right with probability p and to the left with prob $(1 - p)$. His steps are independent. It X denote us position after n steps. Find the distribution $\frac{x+n}{2}$ and then find $E(X)$.

Let, X_i be the direction taken by the drunk at the ith step.

Define $X_i = \begin{cases} +1 & \text{if he takes the ith step to the right} \\ -1 & \text{if he takes the ith step to the left} \end{cases}$

$\Rightarrow \frac{X_i+1}{2} = \begin{cases} 1 & \text{with prob} \quad p \\ 0 & \text{with prob} \quad (1-p). \end{cases} \qquad i = 1, \ldots, n.$

Clearly, $\frac{X_i+1}{2} \overset{iid}{\sim}$ Bernoulli (p), $i = 1(1)n$,

$\Rightarrow \sum_{i=1}^{n} \left(\frac{X_i+1}{2}\right) \backsim \text{Bin}(n, p)$

So, $\sum_{i=1}^{n} X_i$ gives the position of the trunk after nth step. Here, $X = \sum_{i=1}^{n} x_i$ and $\left(\frac{X+n}{p}\right) \backsim \text{Bin}(n, p)$.

$$E\left(\frac{X+n}{2}\right) = np$$
$$\Rightarrow E(X) = 2n\left(p - \frac{1}{2}\right)$$

If $p = \frac{1}{2}, E(X) = 0$ i.e. the expected position of the drunk after the few steps is at the origin.

5. Assume that n random variables $X_1, X_2, \ldots X_n$ are independent and each takes the values +1 and -1 with prob p and q (respectively). Find the distribution of the R.V's. $Y_n = X_1 X_2 \ldots X_n, n \geqslant 1$.

X_n takes only the values ± 1.

$$\begin{aligned} p[Y_n = \pm 1] &= \text{Prob that there are even and of } X_i'\text{s are} - 1 \\ &= p[Z = \text{ even }] \text{ where } Z = \text{ number of X's taking negative value and } \quad Z \sim \text{Bin}(n, q) \\ &= p[Z = 0] + p[Z = 2] + \ldots \\ &= \binom{n}{0} p^0 q^n + \binom{n}{2} p^2 q^{n-2} + \ldots \\ &= \frac{1}{2}\left\{(p+q)^n + (p-q)^n\right\} \\ &= \frac{1}{2}\left\{1 + (p-q)^n\right\} \end{aligned}$$

$$\begin{aligned} (p+q)^n &= \binom{n}{0} q^0 p^n + \binom{n}{1} q p^{n-1} + \binom{n}{2} q^2 p^{n-2} + \ldots \\ (p+q)^n + (p-q)^n &= 2\left\{\binom{n}{0} q^0 p^n - \binom{n}{1} q p^{n-1} + \binom{n}{2} q^2 p^{n-2} - \ldots\right\} \\ \Rightarrow \frac{1}{2}\left\{1 + (p-q)^n\right\} &= \binom{n}{0} q^0 p^n + \binom{n}{2} q^2 p^{n-2} + \cdots \end{aligned}$$

6. A man makes 10 independent attempt at a target and probability of getting the target at each attempt is $\frac{5}{6}$, the conditional probability that his last two attempts are successful given that he has total of 7 successful attempts?

A: 7 successful attempt.

B : Last two attempts are success in 10 independent trails.

Required probability $= P(B \mid A) = \frac{P(A \cap B)}{P(A)}$

$$= \frac{\binom{8}{5}\left(\frac{5}{6}\right)^5 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)\left(\frac{5}{6}\right)}{\binom{10}{7}\left(\frac{5}{6}\right)^7 \left(\frac{1}{6}\right)^3} = \frac{\binom{8}{5}}{\binom{10}{7}}$$

where, $A \cap B =$ The last two attempts are sucessful and in the 1st 8 attempt there are 5 success.

7. Let $X_1, X_2, \ldots X_6$ be independent RV such that $p\left[X_i = -1\right] = p\left[X_i = 1\right], i = 1(1)6$. Find the probability of $\sum_{i=1}^{6} X_i = 4$.

$X_i = \begin{cases} +1 & ,1/2 \\ -1 & ,1/2 \end{cases}$ Binomial to Bernoulli $\frac{X_i+1}{2} = \begin{cases} 1 & ,1/2 \\ 0 & ,1/2 \end{cases}$

$$\begin{aligned} & P\left\{\left(\sum_{i=1}^{6} X_i\right) = 4\right\} \\ =& P\left(2\sum_{i=1}^{6}\left(\frac{X_i}{2} + \frac{1}{2} - \frac{1}{2}\right) = 4\right) \\ =& P(2\,\text{Bin}(6, 1/2) - 6 = 4) \\ =& P(2\,\text{Bin}(6, 1/2) = 10) \\ =& P(\text{Bin}(6, 1/2) = 5) \\ =& \binom{6}{5}\left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) \\ =& \binom{6}{5}\left(\frac{1}{2}\right)^6 \end{aligned}$$

8. 6 identical fair dice are thrown simultaneously, it S denote no of dice showing even no. on their upper faces. The variance of the RV S is 3/4.

9. Let X_i identically and independently distributed Bernoulli (p). Define $S_k = \sum_{i=1}^{K} X_i$.

i) Show that for any $m < n, S_m$ and $S_n - S_m$ are independent.

ii) Show that the conditional distribution of $(S_m|S_n)$ is independent of p.

iii) Find $E\left(\frac{X_1}{S_n}\right)$ for any i, $\left[E\left(\frac{X_i}{S_n}\right)\right]$ and also find $E\left(\frac{S_m}{S_n}\right)$.

For any $m < n$,

$$S_n - S_m = \sum_{k=m+1}^{n} X_i, \quad S_m = \sum_{i=1}^{m} X_i$$

Now, there are no common X_i in between S_m and $S_n - S_m$. So they are independent.

$$\begin{aligned}
P\left[S_m = x \mid S_n = y\right] &= \frac{P\left[S_m = x, S_n = y\right]}{P\left[S_n = y\right]} \\
&= \frac{P\left[S_m = x, S_n - S_m = y - x\right]}{P\left[S_n = y\right]} \\
&= \frac{\binom{m}{x} p^x q^{n-x} \binom{n-m}{y-x} p^{y-x} q^{n-m-y+x}}{\binom{n}{y} p^y q^{n-y}} \\
&= \frac{\binom{n}{x}\binom{n-m}{y-n}}{\binom{n}{y}}, \quad \text{independent of } p \\
&\sim \text{Hyp}(m, n, y)
\end{aligned}$$

$$\begin{aligned}
E\left(\frac{S_n}{S_n}\right) &= 1 \\
E\left(\frac{\Sigma X_i}{S_n}\right) &= 1 \quad \Rightarrow E\left(\sum_{i=1}^{n}\left(\frac{X_i}{S_n}\right)\right) = 1 \\
\Rightarrow \quad \sum_{i=1}^{n} E\left(\frac{X_i}{S_n}\right) &= 1 \quad \text{[they are iid]} \\
\Rightarrow \quad E\left(\frac{X_i}{S_n}\right) &= \frac{1}{n}
\end{aligned}$$

Since X_i's are identically distributed,

$$\begin{aligned}
E\left(\frac{S_m}{S_n}\right) &= E\left(\frac{X_1}{S_n} + \frac{X_2}{S_n} + \frac{X_3}{S_n} + \cdots + \frac{X_m}{S_n}\right) \\
&= \frac{m}{n}
\end{aligned}$$

4 Result

If $X \sim \text{Bin}(n, p)$,

$$Y = (n - X) \sim \text{Bin}(n, q), \quad q = 1 - p$$

$\therefore X =$ no of success in n BernoullI trails with success prob p.

Then, $(n - X)$ is the no of failure in these n trails.

So, if we now think of the success probability as failure probability,$Y \sim \text{Bin}(n, q)$

10. Consider two persons A and B with, $(n + 1)$ unbiased coins and n unbiased coin respectively. If they toss their coins independently, find the probability that -

i) A will get more heads than B.

ii) They will get equal no of head.

$$P(X<Y)+P(X=Y)+P(X>Y)=1$$

$X=$ no of heads obtained by A.
$Y=$ no of heads obtained by B.

$$X\sim \text{Bin}(n+1,1/2)$$
$$(n+1-X)\sim \text{Bin}(n+1,1/2)$$

$$Y\sim \text{Bin}(n,1/2)$$
$$(n-Y)\sim \text{Bin}(n,1/2)$$

$Y\sim \text{Bin}\left(n,\frac{1}{2}\right),\quad X\sim \text{Bin}\left(n+1,\frac{1}{2}\right)$ independently, $\begin{matrix}X & \equiv n+1-X\\ Y & \equiv n-Y\end{matrix}$
$\therefore X$ and $(n+1-X)$ have the same distribution. Y is $(n-Y)$ have the same distribution.

i)

$$\begin{aligned}P[X>Y]&=P[n+1-X>n-Y]\\&=P[1-X>-Y]\\&=P[X-1<y]\\&=P[X<y+1]\\&=P[X\leqslant y]\\&=1-P[X>Y]\\\Rightarrow 2P(X>Y)&=1\\\Rightarrow P[X>Y]&=\frac{1}{2}\end{aligned}$$

ii) $P[X=Y]=P[X-Y=0]$
Now, $X\sim \text{Bin}\left(n+1,\frac{1}{2}\right), Y\sim \text{Bin}\left(n,1/2\right),(n-Y)\sim \text{Bin}\left(n,\frac{1}{2}\right)$ independently

$$(X+n-Y)\backsim \text{Bin}\left(2n+1,\frac{1}{2}\right)$$

$P[X-Y+n=n]=\binom{2n+1}{n}\left(\frac{1}{2}\right)^{2n+1}$
iii) P(X<Y)$=1-\left[\left(\frac{1}{2}\right)+\binom{2n+1}{n}\left(\frac{1}{2}\right)^{2n+1}\right]$.

Remark

If $X\sim \text{Bin}\left(n_1,p\right)$ and $Y\sim \sin\left(n_2,p\right)$, independently.
$X=$ number of success in n_1 Bernoulli trails.
$Y=$ of number of success in n_2 Bernoulli trails.
$(X+Y)$ can be thought of number of success in (n_1+n_2) Bernoulli trail with success prob p.
So, $(X+y)\sim \text{Bin}\left(n_1+n_2,p\right)$.

5 Mode

The value of X for which $P[X=x]$ is maximum ie. the most probable value x of a discrete RV X is called the mode of the discrete R.V X.
Here PMF, $f(x)=\binom{n}{x}p^x\cdot q^{n-x}\quad x=0,1,2,\ldots n.$
Note that,

$$\frac{f(x)}{f(x-1)}=\frac{\binom{n}{x}p^xq^{n-x}}{\binom{n}{x-1}p^{x-1}q^{n-x+1}}$$

$$= \frac{n-x+1}{x}\frac{p}{q}$$

$$= 1 + \frac{(n+1)p - x}{xq} = \begin{cases} > 1 & \text{if } x < (n+1)p \\ = 1 & \text{if } x = (n+1)p \\ < 1 & \text{if } x > (n+1)p \end{cases}$$

Case I $(n+1)p$ is not an integer, $[(n+1)p] = k$ (say).
Here,

$$\frac{f(1)}{f(0)} > 1, \quad \frac{f(2)}{f(1)} > 1, \ldots, \frac{f(k)}{f(k-1)} > 1, \frac{f(k+1)}{f(k)} < 1, \ldots$$

$$\Rightarrow f(0) < f(1) < \ldots < f(k-1) < f(k) > f(k+1) > f(k+2) > \ldots$$

Clearly, $f(k)$ is the maximum hence $k = [(n+1)p]$ is the mode.
Case II $(n+1)p$ is an integer, let $(n+1)p = k$ (say).
Here,

$$\frac{f(1)}{f(0)} > 1, \ldots \quad \frac{f(k-1)}{f(k-2)} > 1, \quad \frac{f(k)}{f(k-1)} = 1. \quad \frac{f(x+1)}{f(x)} < 1 \ldots$$

.

$$\Rightarrow f(0) < f(1) < \ldots < f(k-2) < f(k-1) = f(k) < f(k+1) > \ldots$$

Clearly, $f(k)$ and $f(k-1)$ is the maximum.
Hence $x = np; (n+1)P$ are the modes for Binomial distribution.

Observation

$f(x)$ is an increasing function of x when $x < (n+1)p$ and it is a decreasing function of x when $x > (n+1)p$.
When $(n+1)p$ is an integer. $f(x) = f(x-1)$.
Therefore, $f(x)$ is maximum at $x = (n+1)p$ and $x = (n+1)p - 1$ when $(n+1)p$ is an integer.
But when $(n+1)p$ is not an integer, $f(x)$ has a single maximum at $x = [(n+1)p]$.

Application

1. If $X \sim \sin(10, p)$. Find the restricted range of p for which, there would be a single mode at $X = 1$.

$$\begin{aligned} &[(n+1)p] = 1 \\ \Rightarrow &[11p] = 1 \\ \Rightarrow &1 \leqslant 11p < 2 \\ \Rightarrow &\frac{1}{11} \leqslant p < \frac{2}{11} \end{aligned}$$

2. If a fair coin is tossed $2n$ times. Find the most probable no of head obtained. If m denotes the most probable no of heads, show that

$$\frac{1}{2\sqrt{n}} < P[X = m] < \frac{1}{\sqrt{2n+1}}$$

Here, $\left[(2n+1)\frac{1}{2}\right] = \left[n + \frac{1}{2}\right] = n$.
X = number of heads, $X \sim \text{Bin}\left(2n, \frac{1}{2}\right)$, mode is at n.

$$\begin{aligned} P[X = n] &= \binom{2n}{n}\left(\frac{1}{2}\right)^{2n} = \frac{(2n)!}{n!n!\cdots 2^{2n}} \\ &= \frac{1.3 \cdot 5 \cdots (2n-1)2^n n!}{n!2^n n!2^n} \\ &= \frac{1.3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \end{aligned}$$

Let us consider, a function $g(t) = 1 - \frac{1}{t}, t = 1, 2, \ldots,$
Now, $g(t-1) < g(t) < g(t+1)$
Now, for

$$\begin{aligned} & g(2t-1) < g(2t) < g(2t+1) \\ \Rightarrow & \frac{2t-2}{2t-1} < \frac{2t-1}{2t} < \frac{2t}{2t+1} \\ \Rightarrow & g(2t) < g(2t+1) \\ \Rightarrow & \frac{2t-1}{2t} < \frac{2t}{2t+1} \quad t = 1, 2, \ldots \end{aligned}$$

Putting $t = 1, 2, \ldots, n$

$$\begin{gathered} \frac{1}{2} < \frac{2}{3} \\ \frac{3}{4} < \frac{4}{5} \\ \ldots\ldots \\ \frac{2n-1}{2n} < \frac{2n}{2n+1} \end{gathered}$$

Multiplying all these inequalities,

$$\begin{aligned} & \frac{1 \cdot 3 \ldots (2n-1)}{2 \cdot 4 \ldots 2n} < \frac{2 \cdot 4 \ldots 2n}{1 \cdot 3 \cdot 5 \ldots (2n+1)} \\ \Rightarrow & \frac{\{1 \cdot 3 \ldots (2n-1)\}^2}{\{2 \cdot 4 \ldots 2n\}^2} < \frac{1}{2n+1} \\ \Rightarrow & P[X = n] < \frac{1}{\sqrt{2n+1}} \end{aligned} \tag{1}$$

Now for, $g(2t-1) < g(2t)$

$$\Rightarrow \frac{2t-2}{2t-1} < \frac{2t-1}{2t}$$

Putting $t = 2, 3, \ldots n$

$$\frac{2}{3} < \frac{3}{4}, \frac{4}{5} < \frac{5}{6} \quad \cdots \quad \frac{2n-2}{2n-1} < \frac{2n-1}{2n}$$

$$\Rightarrow \frac{1}{2.2n} < \frac{\{1.3.5 \ldots\ldots (2n-1)\}^2}{(2.4.6 \ldots 2n)^2}$$

$$\Rightarrow \frac{1}{2\sqrt{n}} < P[X = n]. \tag{2}$$

Combining (1) and (2), $\frac{1}{2\sqrt{n}} < P[X = n] < \frac{1}{\sqrt{2n+1}}$
Note, $\lim_{n \to \infty} P[X = n] = 0.$

3. If $X \sim \text{Bin}(n, p)$, show that $P[X = k] < \frac{1}{k - np}$

$$k > np \Rightarrow k \geqslant [np] + 1$$

Now, we know that $f(x)$ is a non-decreasing function of X, when $X \geqslant (n+1)p$
Now, $[np] + 1 \geqslant [(n+1)p]$
$\therefore f(x)$ is a decreasing (non-increasing) function of x when $x > [np] + 1$
Now for, $k > np$

$$\sum_{x=[np]+1}^{k} f(x) < 1 \tag{1}$$

Again,

$$\sum_{x=[np]+1}^{k} f(x) > \sum_{x=[np]+1}^{k} f(k) = f(k)[k-[np]] \tag{2}$$

Using (2) in (1) we get, $1 > f(k)[k-[np]]$

$$\begin{aligned} np &> [np] \\ -np &< -[np] \\ k-np &< k-[np] \\ f(k) &< \frac{1}{k-[np]} < \frac{1}{k-np} \\ k-np &< k-[np] \\ \frac{1}{k-np} &> \frac{1}{k-[np]} \end{aligned}$$

6 Mean Deviation about Mean

Let $X \sim \text{Bin}(n,p), E(X) = np = \mu$.

$$\begin{aligned} MD_{\mu}(x) &= E|x-np| \\ &= \sum_{x=0}^{n} |x-np| f_X(x) \\ &= \sum_{x=0}^{[k]} (np-x) f_X(x) + \sum_{x=[k]+1}^{n} (x-np) f_X(x) \end{aligned}$$

Now, $E(X-\mu) = 0$

$$\begin{aligned} &\Rightarrow \sum_{x=0}^{n} (x-\mu) f_X(x) = 0 \\ &\Rightarrow \sum_{x=0}^{[\mu]} (\mu-x) f_X(x) + \sum_{x=[\mu]+1}^{n} (x-\mu) f_X(x) = 0 \\ &\Rightarrow \sum_{x=0}^{[\mu]} (x-\mu) f_X(x) = \sum_{x=[\mu]+1}^{n} (x-\mu) f_X(x) \end{aligned}$$

Hence,

$$\begin{aligned} MD_{\mu}(x) &= 2\sum_{x=0}^{[\mu]} (\mu-x) f(x) \\ &= 2\sum_{x=0}^{[\mu]} \{np - x(p+q)\} \binom{n}{x} p^x q^{n-x} \\ &= 2\sum_{x=0}^{[\mu]} \{(n-x)p - xq\} \binom{n}{x} p^x q^{n-x} \\ &= 2\left[\sum_{n=0}^{[\mu]} (n-x)p \binom{n}{x} p^x q^{n-x} - \sum_{x=0}^{[\mu]} xq \binom{n}{x} p^x q^{n-x}\right] \end{aligned}$$

Let us take,

$$\begin{aligned} g(x) &= xqf_X(x) \\ &= xq\frac{n!}{(n-x)!x!}p^x q^{n-x} \\ &= \frac{n!}{(n-x)!(x-1)!}p^x q^{n-x+1} \end{aligned}$$

Therefore,

$$g(x+1) = \frac{n!}{(n-x+1)!x!}p^{x+1}q^{n-x} = (n-x)pf_x(x)$$

$$\begin{aligned} \mathrm{MD}_\mu(x) = 2\sum_{n=0}^{[\mu]}[g(x+1)-g(x)] \quad &= 2g[[\mu]+1] \\ = 2[[\mu]+1]qf([\mu]+1) \end{aligned}$$

Note:

$$\frac{f(x+1)}{f(x)} = \frac{\binom{n}{x+1}p^{x+1}q^{n-x-1}}{\binom{n}{x}p^x q^n} = \frac{(n-x)}{(n+1)}\frac{p}{q}$$

$$\Rightarrow q(x+1)f(x+1) = p(n-x)f(x)$$

If $x = [\mu]$, $\Rightarrow \quad q[[u]+1]f([u]+1) = p[X-[u]] = \mathrm{MD}\mu(x)$.

4. Show that for large n,

$$\frac{\mathrm{MD}_\mu(x)}{\mathrm{SD}(x)} \cong \sqrt{\frac{2}{\pi}} \cong \frac{4}{5}$$

For large $n, [np]+1 \approx np$ [integer]
Therefore, MD $\approx 2qnp\mathrm{f}(np)$

$$\therefore 2qnp\binom{n}{np}p^{np}q^{n-np}$$

By Strirlings approximation,

$$\begin{aligned} n! &= \sqrt{2\pi}e^{-n}n^{n+\frac{1}{2}} \\ \mathrm{MD} &= 2qnp\left[\frac{\sqrt{2\pi}e^{-n}n^{n+\frac{1}{2}}p^{np}q^{n-np}}{\left\{\sqrt{2\pi}e^{-np}(np)^{np+\frac{1}{2}}\right\}\sqrt{2\pi}e^{-nq}(nq)^{nq+1/2}}\right] \\ &= 2qnp\left[\frac{n^{n+1/2}p^{np}q^n}{\sqrt{2\pi}n^{n+1}p^{np+\frac{1}{2}}q^{nq+\frac{1}{2}}}\right] \\ &= 2qnp\left[\frac{n^{-1/2}p^{-1/2}q^{-1/2}}{\sqrt{2\pi}}\right] \\ &= \frac{2qnp}{\sqrt{2\pi}\sqrt{npq}} \\ &= \sqrt{\frac{2}{\pi}}\sqrt{npq} \\ \therefore \quad \frac{\mathrm{MD}\mu(x)}{\mathrm{SD}(x)} &= \sqrt{\frac{2}{\pi}} \end{aligned}$$

5. Let $X_n \sim \mathrm{Bin}(n,p)$. Let $\mu_n = E(X_n)$, show that $\mu_n = \mu_{n-1} + p$. Hence find μ_n.

$\{Y_n\}$ be a sequence of iid Bernoulli(p) RV, $X_n \sim \text{Bin}(n,p)$

$$\Rightarrow X_n = \sum_{i=1}^{n} Y_i = X_{n-1} + Y_n$$

where $X_{n-1} = \sum_{i=1}^{n-1} Y_i \sim \text{Bin}(n-1,p)$

$$\begin{aligned}
\Rightarrow E(X_n) &= E(X_{n-1}) + E(Y_n) \\
\Rightarrow \mu_n &= \mu_{n-1} + p \\
&= \mu_{n-2} + 2p \\
&\vdots \\
&= E(X_1) + \overline{n-1}p = np
\end{aligned}$$

6. Let $X \sim \text{Bin}(n,p)$, show that $E\left(\frac{1}{X+1}\right) = \frac{1-q^{n+1}}{(n+1)p}$

$$\begin{aligned}
E\left(\frac{1}{X+1}\right) &= \sum_{x=0}^{n} \frac{1}{x+1} \frac{n!}{x!(n-x)!} p^x q^{n-x} \\
&= \frac{1}{(n+1)p} \sum_{n=0}^{n} \frac{(n+1)!}{(x+1)!(\overline{n+1} - \overline{x+1})!} p^{(x+1)} q^{n-x} \\
&= \frac{1}{(n+1)p} \sum_{y=x+1=1}^{n+1} \binom{n+1}{y} p^y q^{n+1-y} \\
&= \frac{1}{(n+1)p} \left[1 - P(Y=0)\right] \\
&= \frac{1-q^{n+1}}{(n+1)p}
\end{aligned}$$

7 Distribution Function

Let $X \sim \text{Bin}(n,p)$ for an non-negative integer k,

$$\begin{aligned}
P[X \leqslant k] &= \sum_{x=0}^{k} f_X(x) \\
&= \sum_{x=0}^{k} \binom{n}{x} p^x q^{n-x} \\
&= \sum_{x=0}^{k} \binom{n}{x} (1-q)^x q^{n-x} = g(q) \text{ (say). [A continuation function of } q \text{]}
\end{aligned}$$

Differentiating $g(q)$ wrt q,

$$\begin{aligned}
\frac{\partial}{\partial q} g(q) &= \frac{\partial}{\partial q} \left[\sum_{x=0}^{k} \binom{n}{x} (1-q)^x q^{n-x} \right] \\
&= \sum_{x=0}^{k} \binom{n}{x} \left[\frac{\partial}{\partial q} (1-q)^x q^{n-x} \right] \\
&= \sum_{x=0}^{k} \binom{n}{x} \left[(1-q)^x (n-x) q^{n-x-1} - q^{n-x} x (1-q)^{x-1} \right]
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{x=0}^{k} \frac{n!}{(n-x)!x!}(1-q)^x(n-x)q^{n-x-1} - \sum_{x=0}^{k} \frac{n!}{(n-x)!x!}(1-q)^{x-1}xq^{n-x} \\
&= n\left[\sum_{x=0}^{k}\binom{n-1}{x}(1-q)^x q^{n-1-x} - \sum_{x=1}^{k}\binom{n-1}{x-1}(1-q)^{x-1}q^{n-x}\right] \\
&= n\left[\sum_{x=0}^{k}\binom{n-1}{x}(1-q)^x q^{n-x-1} - \sum_{x-1=x'=0}^{k-1}\binom{n-1}{x'}(1-q)^{x'}q^{n-x'-1}\right] \\
&= n\left[\binom{n-1}{k}(1-q)^k q^{n-1-k}\right] \\
&= \frac{n!}{k!(n-1-k)!}(1-q)^k q^{n-1-k} \\
&= \frac{\Gamma n+1}{\Gamma k+1\Gamma n-k}(1-q)^k q^{n-1-k} \\
&= \frac{(1-q)^{\overline{k+1}-1}q^{\overline{n-k}-1}}{\dfrac{\Gamma k+1\Gamma n-k}{\Gamma n-k+k+1}} \\
&= \frac{(1-q)^{k+1-1}q^{n-k-1}}{\beta\{(n-k),(k+1)\}}
\end{aligned}
$$

$$
\begin{aligned}
&\therefore \quad g'(t) = \frac{(1-t)^{\overline{k+1}-1}(t)^{n-k-1}}{\beta(n-k,k+1)} \\
&\Rightarrow \quad \int_0^q g'(t)dt = \int_0^q \frac{(1-t)^{\overline{k+1}-1}t^{n-k-1}}{\beta(n-k,k+1)}dt \\
&\Rightarrow \quad g(q) - g(0) = \frac{\int_0^q (1-t)^{\overline{k+1}-1}t^{n-k-1}}{\beta(n-k,k+1)}dt \\
&\Rightarrow \quad g(q) = \frac{\int_0^q (1-t)^{\overline{k+1}-1}t^{n-k-1}}{\beta(n-k,k+1)}dt \quad \text{as} g(0) = 0
\end{aligned}
$$

Incomplete Beta function is denoted by $I_q(n-k,k+1)$.

$$\therefore P[X \leqslant k] = I_q(n-k,k+1)$$

Since $I_q(n-k,k+1)$ is an increasing function of q, as q increases the upper limit of the integral also increases. Therefore, C.D.F of X is an increasing function of q that is a decreasing function of p.

Tail function is given by,

$$
\begin{aligned}
P[X>k] &= 1 - P[X \leqslant k] \\
&= 1 - I_q(n-k,k+1) \\
P[X>k] &= 1 - \int_0^q \frac{(1-t)^{\overline{k+1}-1}(t)^{n-k-1}dt}{\beta(n-k,k+1)} \\
&= \frac{\int_0^1 (1-t)^{k+1-1}(t)^{n-k-1}dt - \int_0^q (1-t)^{k+1-1}t^{n-k-1}dt}{\beta(n-k,k+1)} \\
&= \frac{\int_q^1 (1-t)^{k+1-1}t^{n-k-1}dt}{\beta(n-k,k+1)} \\
&= \frac{\int_0^p z^{\overline{k+1}-1}(1-z)^{n-k-1}dz}{\beta(n-k,k+1)}, \quad z = 1-t \\
&= I_p(k+1,n-k)
\end{aligned}
$$

1. Consider two random variable, X_1, and X_2 such that $X_1 \sim \text{Bin}(p_1)$ and $X_2 \sim \text{Bin}(n, p_2)$ where $p_1 > p_2$. Show that, for any non-negative integer $k, P[X_1 \leqslant k] < P[X_2 \leqslant k]$.

$$\begin{aligned} q_i &= 1 - p_i \quad i = 1, 2 \quad q_1 < q_2 \\ P[X_1 \leqslant k] &= I_{q_1}(n-k, k+1) \\ P[X_2 \leqslant k] &= I_{q_2}(n-k, k+1) \end{aligned}$$

Now, $\quad I_{q_1}(n-k, k+1) < I_{q_2}(n-k, k+1)$

$$P[X_1 \leqslant k] < P[X_2 \leqslant k]$$

2. If $X \sim \text{Bin}(n, p)$. Show that $P[X \geqslant k] < \binom{n}{k} p^k$.

$$\begin{aligned} p[X \geqslant k] =& p[X > k-1] \\ =& I_p(k, n-k+1) = \frac{\int_0^p t^{k-1}(1-t)^{n-k} dt}{\beta(k, n-k+1)} \\ \leqslant& \frac{\int_0^p t^{k-1} dt}{\beta(k, n-k+1)} \leqslant \frac{p^k \Gamma n + 1}{k \Gamma k \Gamma n - k + 1} \\ \leqslant& \binom{n}{k} p^k \end{aligned}$$

3. A coin with probability of head p is tossed n times where n is an odd positive integer. Let X denotes the no of head obtained. Y denotes the no of tail obtained in n throws. If $P[X > Y] = P[X < Y]$, show that the coin is fair.

X = number of head obtained.

Y = number of tails obtained.

$$\begin{aligned} \therefore P[X > Y] &= P[X > n - X] = P[2X > n] \\ &= P[X > n/2] = P\left[X \geqslant \left[\frac{n}{2}\right] + 1\right] = P\left[X > \left[\frac{n}{2}\right]\right] \\ &= I_p\left[\left[\frac{n}{2}\right] + 1, n - \left[\frac{n}{2}\right]\right] \end{aligned}$$

$$\begin{aligned} P[X < Y] &= P[X < n - X] \\ &= P\left[X < \frac{n}{2}\right] = P\left[X \leqslant \left[\frac{n}{2}\right]\right. \\ &= I_q\left[n - \left[\frac{n}{2}\right], \left[\frac{n}{2}\right] + 1\right] \\ &= I_p\left[\left[\frac{n}{2}\right] + 1, n - \left[\frac{n}{2}\right]\right] = \int_0^p \frac{t^{m-1}(1-t)^{n-1}}{\beta(m, n)} dt \\ &= I_q\left[n - \left[\frac{n}{2}\right], \left[\frac{n}{2}\right] + 1\right] \end{aligned}$$

4. $X \sim \text{Bin}(n, p)$, show that,

i) $P[X \geqslant r] \leqslant \dfrac{rq}{(r - np)^2}, \quad r > np.$

ii) $P[X \leqslant r] \leqslant \dfrac{(n-r)p}{(np - r)^2}, \quad r < np.$

i)

$$\begin{aligned} & P[X \geqslant r] \quad \text{for } r > np \\ = \; & P[X - np \geqslant r - np] \quad \text{for } (r - np) > 0 \\ = \; & P[(X - np)^2 \geqslant (r - np)^2] \quad \text{for } (X - np) > 0 \end{aligned}$$

$$
\begin{aligned}
&\leqslant P\left[(X-np)^2 \geqslant (r-np)^2\right] \\
&\leqslant \frac{E(X-np)^2}{(r-np)^2} \\
&= \frac{npq}{(r-np)^2} \leqslant \frac{rq}{(r-np)^2}, \quad \text{as } r > np
\end{aligned}
$$

ii) For $r < np$,

$$
\begin{aligned}
&\Rightarrow r < n - nq \\
&\Rightarrow nq < (n-r) \\
P[X \leqslant r] = p = P[-X \geqslant -r] &= P[np - X \geqslant np - r] \\
&\leqslant P\left[(np-X)^2 \geqslant (np-r)^2\right] \\
&\leqslant \frac{npq}{(np-r)^2} \\
&\leqslant \frac{(n-r)p}{(np-r)^2}
\end{aligned}
$$

5. Show that, Var $(X) \leqslant \dfrac{n}{4}$.

$$
\begin{aligned}
\text{Var } (X) &= np(1-p) \\
&= np - np^2 \\
&= -n\left(p^2 - p\right) \\
&= -n\left[p^2 - 2p \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \frac{1}{4}\right] \\
&= -n\left[\left(p - \frac{1}{2}\right)^2 - \frac{1}{4}\right] \\
&= \frac{n}{4} - n\left(p - \frac{1}{2}\right)^2 \leqslant n/4
\end{aligned}
$$

8 Recursion Relation of Binomial Distribution

$$
\begin{aligned}
\mu_r = E[X - E(X)]^r &= E[X - np]^r \\
&= \sum_{x=0}^{\infty} (x-np)^r \binom{n}{x} p^x q^{n-x}
\end{aligned}
$$

Differentiating w.r.t to p, we get,

$$
\frac{\partial u_r}{\partial p} = \sum_{x=0}^{n} \binom{n}{x} r(x-np)^{r-1}(-n)p^x q^{n-x} + \sum_{x=0}^{n} (x-np)^n \binom{n}{x} x p^{x-1} q^{n-x} + \sum_{x=0}^{n} (x-np)^n \binom{n}{x} p^x (n-x) q^{n-x-1}(-1)
$$

$$
= -nr \sum_{x=0}^{n} \binom{n}{x} (x-np)^{n-1} p^x q^{n-x} + \sum_{x=0}^{n} (x-np)^r \binom{n}{x} p^x q^{n-x} \left\{\frac{x}{p} - \frac{n-x}{q}\right\}
$$

$$
= -nr\mu_{r-1} + \sum_{x=0}^{n} (x-np)^n \binom{n}{x} p^x q^{n-x} \left(\frac{xq - np + xp}{pq}\right)
$$

$$
= -nr\mu_{n-1} + \frac{1}{pq} \sum_{x=0}^{n} (x-np)^{n+1} \binom{n}{x} p^x q^{n-x}
$$

$$\frac{\partial \mu_n}{\partial p} = -nr\mu_{n-1} + \frac{1}{pq}\mu_{n+1}$$
$$\Rightarrow \quad \frac{1}{pq}\mu_{n+1} = \left(nr\mu_{n-1} + \frac{\partial \mu_n}{\partial p}\right)$$
$$\Rightarrow \quad \mu_{n+1} = pq\left(nr\mu_{n-1} + \frac{\partial u_n}{\partial p}\right)$$

This is called recursion relation of Binomial Distribution.
$\mu_0 = 1, \quad \mu_1 = 0.$
$\mu_2 = pq\left[n\mu_0 + \frac{\partial \mu_1}{\partial p}\right] = npq.$
$\mu_3 = pq\left[(2n)\mu_1 + \frac{\partial u_2}{\partial p}\right] = pq[-np + nq] = pqn(q-p) = npq(q-p).$

$$\begin{aligned}
\mu_4 &= pq\left[(3n)\mu_2 + \frac{\partial \mu_3}{\partial p}\right] \\
&= pq\left[3n^2pq + nq(q-p) + np(-1)(q-p) + npq(-1) - 1\right)\Big] \\
&= 3n^2p^2q^2 + npq\left[q^2 - pq - pq + p^2 - pq - pq\right] \\
&= 3n^2p^2q^2 + npq\left[(p+q)^2 - 2pq - 4pq\right] \\
&= 3n^2p^2q^2 + npq[1 - 6pq]
\end{aligned}$$

Skewness $= \dfrac{\mu_3}{\mu_2^{3/2}} = \dfrac{npq(q-p)}{(npq)^{3/2}} = \dfrac{npq(q-p)}{npq\sqrt{npq}} = \dfrac{q-p}{\sqrt{npq}}.$
If $\dfrac{q-p}{\sqrt{npq}} = 0, \quad \Rightarrow p = q = 1/2 \quad \Rightarrow$ symmetric about $1/2$.
If $p < 1/2 \quad \Rightarrow$, positively skewed.
If $p > 1/2 \quad \Rightarrow$, negatively skewed.

$$\begin{aligned}
\text{kurtosis} &= \frac{\mu_4}{\mu_2^2} - 3 \\
&= \frac{3n^2p^2q^2 + npq[1-6pq]}{n^2p^2q^2} - 3 \\
&= \frac{3n^2p^2q^2 + npq[1-6pq] - 3n^2p^2q^2}{n^2p^2q^2} = \frac{(1-6pq)}{npq}
\end{aligned}$$

If, $\dfrac{1-6pq}{npq} = 0 \Rightarrow 6pq = 1 \Rightarrow pq = 1/6.$
$pq = 1/6 \quad \Rightarrow$ mesokurtic.
$pq < 1/6 \quad \Rightarrow$ leptokurtic.
$pq > 1/6 \quad \Rightarrow$ pletykurtic.

9 Mean Deviation about mean

$$
\begin{aligned}
\mathrm{MD}_\mu &= E\{|X-np|\}. \\
&= \sum_{x=0}^{n} |x-np| \binom{n}{x} p^x q^{n-x}. \\
&= 2\sum_{x=k+1}^{n} (x-np) \binom{n}{x} p^x q^{n-x}. \\
&= 2\sum_{x=k+1}^{n} \{x(p+q)-np\} \binom{n}{x} p^n q^{n-x}. \\
&= 2\sum_{x=k+1}^{n} \{xq-(n-x)p\} \frac{n!}{x!(n-x)!} p^x q^{n-x}. \\
&= 2\sum_{x=k+1}^{n} \left\{ \frac{n!}{(x-1)!(n-x)!} p^x q^{n-x+1} - \frac{n!}{x!(n-x-1)!} p^{x+1} q^{n-x} \right\} \\
&= 2\sum_{x=k+1}^{n} \{v_x - v_{x+1}\}. \\
&= 2\left\{ v_{k+1} - v_{k+2} + v_{k+2} - v_{k+3} + \cdots - \underbrace{v_{n+1}}_{=0} \right\} \\
&= 2v_{k+1} \\
&= 2\frac{n!}{(k+1)!(n-k-1)!} p_0^{k+1} q^{n-k} \\
&= 2npq\frac{(n-1)!}{k!(n-k-1)!} p^k q^{n-1-k} \\
&= 2npq\binom{n-1}{k} p^k q^{n-1-k}
\end{aligned}
$$